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EDUCATION

M.Ed. in Learning Design & Technologies | (Expected May 2026)*

University of South Carolina, Columbia, South Carolina

Focus: Instructional systems design, Adult Learning Theory

M.S. in Computer Science: Interactive Intelligence

Georgia Institute of Technology, Atlanta, Georgia

Focus: AI Ethics & Society, Human-Computer Interaction, Health Informatics, Cognitive Science

M.B.A. in Strategic Innovation

Georgia Institute of Technology, Atlanta, Georgia

Focus: Technology strategy, product development, innovation management

M.S. in Human-Computer Interaction

Iowa State University, Ames, Iowa

Focus: User-centered design, cognitive engineering

M.S. in Entrepreneurship

University of Florida, Gainesville, Florida

Focus: Technology commercialization, venture creation

B.G.S. in Web Technologies

University of Connecticut, Storrs, Connecticut

RESEARCH INTERESTS

Primary Research Areas:

Explainable AI for architectural decision support, software engineering economics, empirical evaluation of AI systems in enterprise contexts, trustworthy AI deployment

Secondary Research Areas:

Multi-agent reinforcement learning, human-AI collaboration, cloud-native architecture evaluation, AI governance frameworks, value realization in enterprise AI adoption

ACADEMIC RESEARCH EXPERIENCE

Graduate Research Collaborator | 2023 - 2024

Vanderbilt University, MCPsych Leap Laboratory, Nashville, Tennessee

Research Advisor: [Add name if permitted]

- Conducted empirical research applying regression models and signal processing techniques to multimodal psychophysiological datasets for mental health applications
- Developed cloud-native research data pipelines integrating real-time ingestion, transformation, and statistical validation protocols
- Implemented probabilistic modeling and anomaly detection algorithms improving research data quality and measurement validity
- Collaborated with interdisciplinary research team (clinical psychology, data science), translating theoretical constructs into computational implementations

Independent Research | 2020 - Present

Enterprise AI Adoption & ROI Measurement

Location: Atlanta, Georgia

- Developed empirically-grounded ROI measurement framework for enterprise AI adoption, addressing gaps in software engineering economics literature regarding value realization in agentic AI systems
- Conducted field research with 15+ Fortune 500 organizations implementing

multi-agent AI, identifying systematic barriers to adoption and measurement challenges in production environments

- Created operationalized metrics linking architectural decisions to business value outcomes (cost efficiency, delivery velocity, security posture), validated across heterogeneous enterprise contexts
- Research findings inform industry best practices for trustworthy AI deployment and responsible AI governance frameworks.
- Empirical studies on LLM safety evaluation methods across multiple model architectures (GPT-4, Claude, Llama family), examining hallucination detection, toxicity mitigation, and adversarial robustness
- Developed automated testing pipelines for AI safety evaluation integrated into production ML workflows

Learning Analytics Research | 2016 - 2018

Trilogy Education Services (2U), Atlanta, Georgia

- Conducted applied research on student behavior modeling using natural language processing and behavioral trace data from GitHub APIs to predict at-risk learners in online coding education
 - Developed sentiment analysis prototypes, processing student feedback to identify engagement patterns and intervention opportunities
 - Created data-driven recommendation engine providing personalized learning interventions based on individual learning trajectories
 - Research resulted in measurable learning outcomes: 28% increase in student engagement, 15% improvement in assignment completion rates
 - Findings informed instructional design practices across 15+ university partners
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Technical Publications & Professional Reports

- **"Trustworthy AI Implementation Frameworks"** - Comprehensive technical guides integrating fairness testing, bias mitigation, and explainability into production ML systems. Adopted by 15+ enterprise partners across fintech, healthcare, and manufacturing sectors. Industry Technical Report Series, 2023-Present.
- **"Enterprise AI Platform Reference Architectures"** - Published deployment

guides and best practices documentation (10,000+ combined views). Topics: Large language model framework integration, robotics simulation workflows, AI microservices architecture, GPU-accelerated distributed training. Industry Technical Documentation, 2022-Present.

- **"AWS Machine Learning Essentials Lab Series"** - Authored hands-on technical learning modules completed by 10,000+ practitioners (4.5/5.0 average rating). Topics: SageMaker model training, inference optimization, MLOps pipeline design. Amazon Web Services Education, 2020-2022.

Conference Presentations & Invited Talks

- **"Hidden Careers in Computer Science"** - Invited Panelist, Lunch & Learn Series, Georgia Institute of Technology, October 29, 2025
- **"AI in Academic and Industry Contexts"** - Invited Panelist, AI @ UGA Panel, University of Georgia, October 10, 2025
- **"Game Design and Development Career Pathways"** - Invited Panelist, CTRL + CM Game Design Panel, Georgia Institute of Technology, September 29, 2025
- **"Technology Innovation for Emerging Careers"** - Invited Speaker, 8th Annual Upward Bound College & Career Day, Georgia State University, July 2020
- **"Serverless Applications with AWS Lambda"** - Technical Workshop Presenter, HackGT 5, Georgia Institute of Technology, October 2018
- **"Career Pathways in Technology"** - Invited Panelist, General Assembly Career Focus Series, June 2016

Additional Technical Presentations: 25+ technical workshops and presentations (2014-2020) on software development methodologies, AI/ML implementation, and technology innovation at academic and industry venues

Open Source Contributions

- **BootCampSpot v2 Learning Management System** - Lead architect and developer for RESTful API infrastructure and microservices platform serving 50,000+ students across 15+ university partners (UCLA, Georgia Tech, UC Berkeley, Rutgers, Northwestern, UNC, University of Texas Austin). Technologies: Node.js, PostgreSQL, microservices architecture. Trilogy Education Services, 2016-2017.
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TEACHING EXPERIENCE

Lecturer, Business Programming (MGT 3745) | 2019 - Present

Georgia Institute of Technology, Scheller College of Business, Atlanta, Georgia

- Teach 300+ graduate and undergraduate students annually covering Python programming, JavaScript, ML API integration, cloud workflows, and responsible AI frameworks
- Design complete course infrastructure including lectures, hands-on laboratories, industry-relevant projects, and automated grading systems providing immediate detailed feedback
- Integrate cutting-edge AI technologies (retrieval-augmented generation, LLM APIs, prompt engineering) reflecting current industry practices and research developments
- **Research-informed pedagogy:** Applied learning analytics principles to course redesign based on student performance data and feedback, resulting in 12-point average grade improvement and student satisfaction increase from 4.2 to 4.7/5.0
- **Impact on student outcomes:** 85% of students reported course directly contributed to career advancement opportunities in technology roles
- Mentor students on technical career transitions, graduate school applications, and research opportunities

Section Teacher, CS106A Programming Methodologies | 2020 - 2022

Stanford University, Computer Science Department, Stanford, California

- Facilitated interactive Python programming sections for 30-40 students per session in foundational computer science course during pandemic transition to remote learning
- Developed supplementary instructional materials and practice problems addressing common conceptual difficulties identified through formative assessment
- Provided individualized feedback and mentoring during office hours on algorithmic thinking and programming fundamentals
- Received consistently positive student feedback for creating inclusive learning environment and delivering accessible explanations of complex concepts

Lead Instructor, Data Science & Analytics Bootcamp | 2020

University of California Berkeley Extension, Berkeley, California

- Designed and delivered intensive 24-week data science curriculum for career-transitioning professionals
- Taught comprehensive curriculum: Python programming, statistical analysis, machine learning fundamentals, production ML workflows, cloud deployment
- Created authentic hands-on projects simulating real-world data science work: exploratory data analysis, predictive modeling, ML system deployment
- Emphasized reproducible research practices, version control, and collaborative development workflows
- Mentored students through capstone projects involving end-to-end ML pipeline development

Instructor, Full-Stack Web Development Bootcamp | 2016 - 2018

Trilogy Education Services (Georgia Tech Partnership), Atlanta, Georgia

- Delivered 24-week intensive coding bootcamp curriculum for career changers entering software development
- Taught full-stack development: HTML/CSS, JavaScript, React, Node.js, databases, deployment
- Integrated agile methodologies and industry coding practices throughout curriculum
- Provided individualized mentorship through office hours addressing diverse learning needs
- Student outcomes: 82% job placement rate within 6 months of graduation

Instructor & Curriculum Designer | 2016 - 2019

General Assembly, Atlanta, Georgia

- Designed and taught JavaScript Development, Front-End Web Development, and technical workshop courses
- Created instructional materials incorporating adult learning theories and evidence-based pedagogical practices

- Served as subject matter expert for circuit-wide course instructional design initiatives
 - Facilitated learning through virtual office hours and asynchronous learning environments
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ACADEMIC SERVICE & LEADERSHIP

Curriculum Advisory Roles

- **Advisory Council Member, Technology Programs** - Penn Foster, 2019-Present
Provide guidance on curriculum design, industry alignment, and emerging technology integration
- **Advisory Council Member, Technology Programs** - Ashworth College, 2016-Present
Advise on program development, learning outcomes assessment, and career preparation strategies
- **Curriculum Advisor, Software Development Certificate** - University of Washington, 2020-Present
Review course content, recommend industry-relevant updates, assess learning pathway effectiveness

Student Mentorship & Outreach

- **Technology Innovation Day Panelist** - Meadowcreek High School, 2016-2018
Presented career pathways in technology, software development, and AI/ML to high school students
- **Career Day Speaker** - Georgia State University Upward Bound Program, 2014-2020
Mentored underrepresented students on STEM career opportunities and academic preparation
- **Fellowship Selection Committee Member** - Code2040, 2017
Evaluated applications for competitive fellowship program supporting Black and Latinx technologists
- **STEM Instructor** - NY Trio Upward Bound Math & Science Program, Morehouse College, 2016

Taught technical workshops to college-bound students from underrepresented backgrounds

Professional Service

- Technical workshop presenter at student hackathons and developer conferences (HackGT, ATL Hoops Hackathon, AT&T Mobile App Hackathon)
 - Subject matter expert reviewer for educational technology curricula at multiple institutions
 - Volunteer instructor for community coding workshops and technology literacy programs
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PROFESSIONAL RESEARCH & DEVELOPMENT EXPERIENCE

Senior Solutions Architect, AI Enterprise | 2022 - Present

Leading GPU/AI Technology Company, Atlanta, Georgia

- Lead enterprise adoption research and technical enablement for AI platform solutions, large language model frameworks, robotics simulation tools, 3D collaboration platforms, and AI microservices across Fortune 500 clients and academic research institutions
- Conduct applied research on GPU-accelerated ML workflows, distributed training architectures, and scalable inference deployment patterns
- Deliver technical training to 500+ partner engineers through 30+ workshops covering production AI system design, trustworthy AI implementation, and performance optimization
- Develop technical enablement content including SDK guides, API documentation, reference architectures, and integration tutorials adopted by enterprise partners
- Collaborate with OEM partner product teams on AI solution roadmaps, translating cutting-edge GPU architecture innovations into customer-facing solutions
- **Research specialization:** Multi-agent AI systems, trustworthy AI deployment frameworks, human-in-the-loop workflows for enterprise production environments

Cloud Technical Curriculum Developer | 2020 - 2022

Amazon Web Services, Seattle, Washington (Remote: Atlanta, Georgia)

- Authored hands-on technical laboratories and tutorials for AWS services: S3, Lambda, serverless architectures, SageMaker, and ML deployment pipelines
- Designed progressive learning pathways from foundational cloud concepts through advanced ML engineering reflecting educational scaffolding principles
- Developed assessment strategies and coding exercises measuring actual skill acquisition aligned with Bloom's taxonomy
- Created AWS ML Essentials lab series recognized as top-rated learning resource: 10,000+ completions, 4.5/5.0 average rating
- Established content quality standards and peer review processes adopted across curriculum team

Technology Innovation Senior Principal | 2018 - 2020

Accenture Innovation Studio (Liquid Studio), Atlanta, Georgia

- Led rapid prototyping and R&D initiatives for AI/ML solutions across Fortune 500 clients in fintech, supply chain logistics, and cybersecurity domains
- Conducted applied research on anomaly detection, AI-driven simulation, and big data visualization
- Developed proof-of-concept applications demonstrating business value of emerging technologies, resulting in \$2M+ in project expansions
- **Featured research project:** Multi-agent drone swarm control interfaces (recognized in 2020 Accenture North America Innovation Yearbook)
- **Impact:** AI-powered anomaly detection system reducing manufacturing defect rates by 22%, deployed across five production facilities
- Mentored junior engineers on ML implementation, cloud architecture patterns, and agile development practices

Senior Software Engineer | 2016 - 2018

Trilogy Education Services (2U), Atlanta, Georgia

- Architected and developed BootCampSpot v2 learning management system

- using RESTful APIs, microservices architecture, and CI/CD pipelines
- Built data-driven learning analytics tools: NLP-based sentiment analysis, GitHub activity tracking, predictive modeling for at-risk student identification
- Implemented platform features improving measurable learning outcomes: 28% increase in student engagement, 15% improvement in assignment completion
- Collaborated with curriculum team identifying platform enhancements based on learning analytics research findings

Selected Prior Experience | 2002-2016

- **Lead Mobile Application Architect** - Enterprise clients including Mercedes-Benz, Macy's, State Farm, IHG (2012-2014)
 - **Lead Consultant, Mobile Solutions Group** - Accenture: SOA architecture for Halliburton, Level 3 Communications, Lowe's (2011-2012)
 - **Product Developer** - NBA Digital/Turner Sports: NBA.com and PGA Tour mobile/tablet applications (2009-2011)
 - **Senior Software Architect** - Anthem Innovation Studio: Healthcare technology prototyping (2016)
 - **Founder & Technical Product Manager** - Base10 Software LLC: Mobile software consultancy (2013-2016)
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HONORS & AWARDS

- **Accenture North America Innovation Yearbook Recognition** - Featured for IoT Drone Swarm Technology Prototype demonstrating multi-agent coordination (February 2020)
- **Hackathon Competition Awards** (Selected)
 - ATL Hoops Hackathon Winner - "Taunt" mobile iOS application (March 2013)
 - City of Atlanta "Govathon" Winner - "Crime Blotter" iOS app for Atlanta Police Department (February 2013)
 - AT&T Mobile App Hackathon Top Winner - Best NFC Gaming Application (March 2013)
 - 22squared Mobilize Hackathon Winner - Team Rocket Jeep iOS application (November 2012)

- **The Black Ball (London) Recognition** - Mobile application design for Keep A Child Alive charity (January 2010)
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TECHNICAL COMPETENCIES

Artificial Intelligence & Machine Learning

- **Large Language Models:** GPT-4, Claude, Llama family, Ollama, prompt engineering, retrieval-augmented generation (RAG)
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, SciPy, Matplotlib, Seaborn, Jupyter
- **AI Safety & Governance:** LlamaGuard, Aegis, Garak, fairness metrics, bias detection, SHAP/LIME explainability
- **Specialized ML:** Computer vision, natural language processing, reinforcement learning, AutoML

Cloud & Distributed Systems

- **Cloud Platforms:** AWS (SageMaker, Lambda, S3, EC2, Batch), Azure (ML Studio, Cognitive Services), GCP (Vertex AI, Cloud Run)
- **Infrastructure as Code:** Docker, Kubernetes, Terraform, Helm
- **Distributed Systems:** Microservices architecture, RESTful APIs, GraphQL, event-driven architectures, message queuing
- **MLOps:** CI/CD pipelines (Jenkins, GitHub Actions), model versioning, A/B testing, monitoring

Programming & Software Engineering

- **Expert Proficiency:** Python, JavaScript/TypeScript, SQL
- **Proficient:** Go, Rust, Swift, Kotlin, R, HTML/CSS
- **Frameworks:** React, Angular, Vue, Node.js, Express, FastAPI, Flask
- **Databases:** PostgreSQL, MySQL, MongoDB, Redis, DynamoDB, CosmosDB
- **Version Control:** Git, GitHub, GitLab, code review practices

Research Methods & Analytics

- **Statistical Analysis:** Regression modeling, hypothesis testing, ANOVA, time series analysis
- **Experimental Design:** A/B testing, randomized controlled trials, quasi-experimental methods
- **Data Analytics:** R, Tableau, statistical modeling, data visualization
- **Qualitative Methods:** Thematic analysis, user interviews, observational studies

Specialized Technologies

- **IoT & Edge Computing:** MQTT, SSE, WebSockets, Raspberry Pi, Arduino, edge AI development platforms
 - **UX Research & Prototyping:** Adobe XD, InVision, Axure, Sketch, Figma, user testing protocols
 - **Learning Technologies:** Canvas LMS, Blackboard, Articulate Storyline, Adobe Captivate, H5P
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PROFESSIONAL CERTIFICATIONS

- **Artificial Intelligence Governance Professional (IAPP AIGP)** - International Association of Privacy Professionals
 - **AWS Certified Solutions Architect - Associate** - Amazon Web Services
 - **Microsoft Certified: Azure AI Fundamentals** - Microsoft
 - **Certified Scrum Product Owner (CSPO)** - Scrum Alliance
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PROFESSIONAL MEMBERSHIPS

- **Association for the Advancement of Artificial Intelligence (AAAI)**
- **Association for Computing Machinery (ACM)**
 - Special Interest Groups: SIGAI, SIGCHI, SIGCSE, SIGSOFT
- **International Association of Privacy Professionals (IAPP)**
- **Institute of Electrical and Electronics Engineers (IEEE)**

- **Interaction Design Foundation (IDF)**
 - **International Association of Innovation Professionals (IAOIP)**
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LANGUAGES

- **English** - Native proficiency